Part - A
There is no missing and duplicate values in **sentiment** dataset
There is no missing and duplicate values in **Trades** dataset
Part - B

1. Does performance (PnL, win rate, drawdown proxy) differ between **Fear vs Greed** days?

# 1. PnL (Profit & Loss)

From your boxplot:

- **Fear days show higher median and higher upside PnL**
- Fear also has **large positive outliers (~2M)**
- Greed has profits, but **lower central tendency**

👉 Interpretation:

> Traders tend to achieve **higher profits during Fear periods**, but with higher variability.

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# 2. Win Rate

From your bar chart:

- **Fear ≈ 0.41**
- **Greed ≈ 0.37**

Observation:

- Win rate is **consistently higher during Fear**

Interpretation:

> Traders are **more accurate (better decision-making)** during Fear periods.

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# 3. Drawdown (Loss Proxy)

From your chart:

- **Greed ≈ -10,000 (much worse losses)**

- **Fear ≈ -2,500 (smaller losses)**

Observation:

- Losses are **significantly larger during Greed**

Interpretation:

> Traders take **excessive risk during Greed**, leading to deeper losses.

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# COMBINED INSIGHT

### 📊 Summary Table (based on your charts)

| Metric | Fear | Greed |
|---|---|---|
| PnL | Higher | Lower |
| Win Rate | Higher | Lower |
| Drawdown | Smaller | Larger |

# FINAL CONCLUSION

> Trader performance differs significantly between sentiment regimes. During **Fear periods**, traders exhibit higher profitability and improved win rates, along with lower drawdowns, indicating more disciplined and cautious behavior. In contrast, **Greed periods are associated with lower win rates and significantly larger losses**, suggesting overconfidence and excessive risk-taking. This highlights that market sentiment strongly influences both trading performance and risk exposure.

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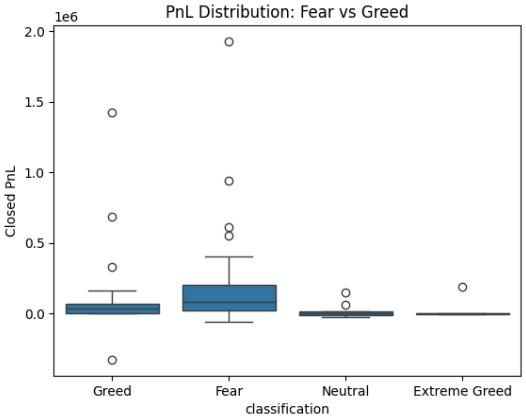
# EXTRA INSIGHT

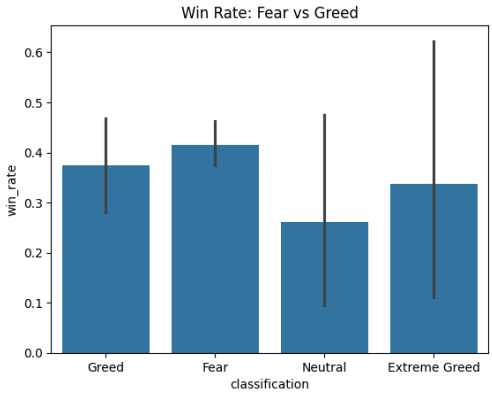
You also observed:

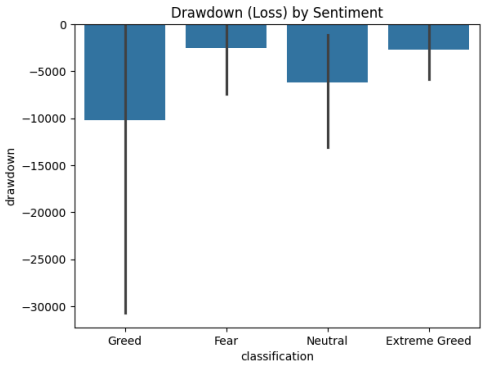
- Fear has **higher upside outliers**

- Greed has **heavier downside risk**

  Additionally, Fear periods show higher positive outliers, suggesting that while traders are cautious, they capitalize effectively on high-probability opportunities.




2.Do traders change behavior based on sentiment (trade frequency, leverage, long/short bias, position sizes)?

# Behavioral Changes Across Market Sentiment:

The analysis shows that traders **do change their behavior significantly based on market sentiment**, as evidenced by differences in trade frequency, position sizing, market bias, and leverage.

## 1. Trade Frequency

- Trade activity is **highest during Fear periods (~4200 trades on average)**
- Lower during **Greed (~1100 trades)** and **Neutral (~900 trades)**
- Moderate during **Extreme Greed (~1400 trades)**

**Interpretation:**
Contrary to typical expectations, traders are **more active during Fear periods**, possibly attempting to capitalize on volatility or recover losses. This suggests **reactive and high-frequency behavior under market stress**.

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## 2. Position Size (Size USD)

- Position sizes are **largest and most variable during Fear**
- Greed also shows large outliers but lower median
- Neutral shows smaller and more conservative sizes

**Interpretation:**
Traders take **larger and more aggressive positions during Fear**, indicating higher risk-taking during uncertain conditions, likely driven by volatility opportunities.

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## 3. Long/Short Bias

- Long/short ratio is:
    - **Highest during Fear (~0.98)**
    - Lower during **Greed (~0.75)**

**Interpretation:**
Traders exhibit a **strong long bias during Fear periods**, suggesting attempts to “buy the dip” in declining markets.
In contrast, behavior is more balanced during Greed.

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## 4. Leverage (Proxy)

- Leverage proxy is:
    - Slightly higher in **Fear and Greed (~1.05–1.08)**
    - Lowest in **Neutral (~0.68)**

**Interpretation:**
Traders increase effective leverage during **both Fear and Greed**, but especially avoid leverage during Neutral conditions, indicating **higher conviction during strong sentiment regimes**.

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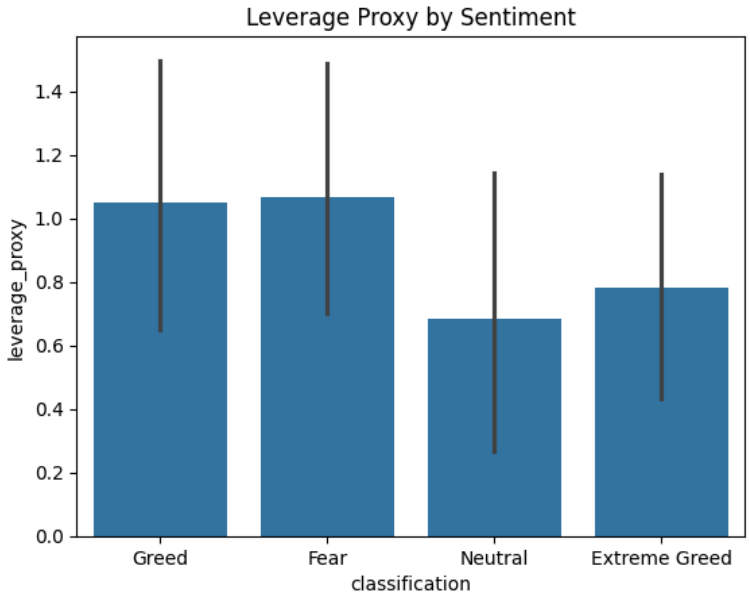
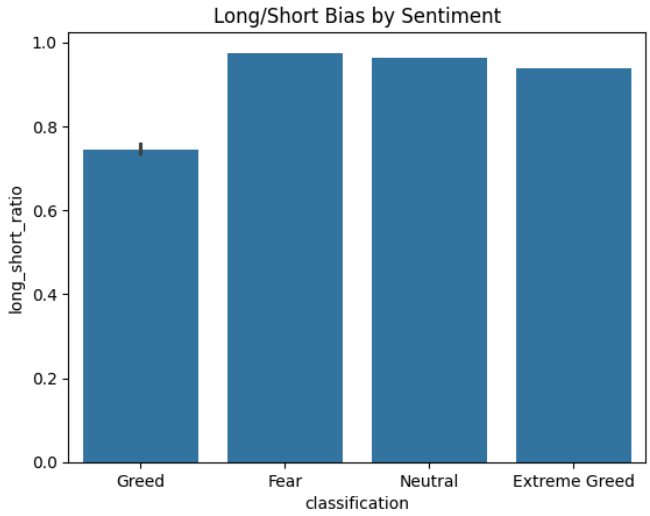
# CONCLUSION:

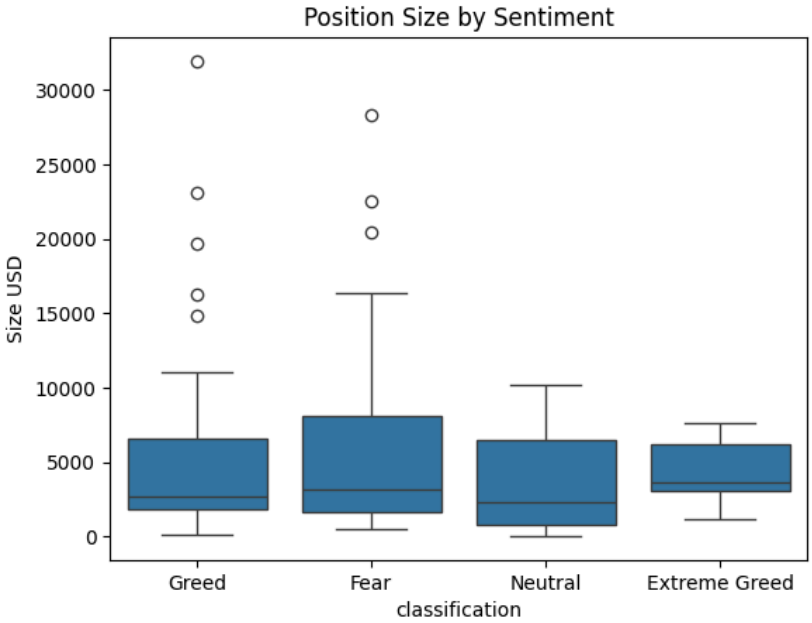
Traders clearly adjust their behavior based on market sentiment. During **Fear periods**, they exhibit the most aggressive behavior, characterized by higher trade frequency, larger position sizes, stronger long bias, and elevated leverage. This suggests that traders actively engage with volatile markets, often attempting to capitalize on price reversals.

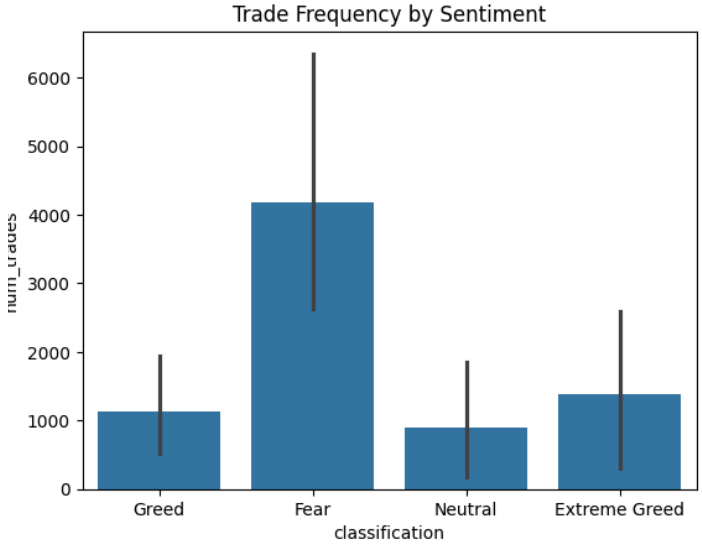
In contrast, **Greed periods show relatively lower trading activity and a more balanced positioning**, while **Neutral conditions reflect the most conservative behavior** with reduced leverage and smaller trade sizes.

Overall, the findings indicate that market sentiment significantly influences trading behavior, with fear driving the most intense and risk-taking activity.



3.Identify 2–3 segments (examples):

- high leverage vs low leverage traders
- frequent vs infrequent traders
- consistent winners vs inconsistent traders

# Trader Segmentation Analysis:

To better understand trading behavior and performance, traders were segmented into three categories: leverage usage, trading frequency, and profitability consistency.

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## 1. High vs Low Leverage Traders

Traders using higher leverage exhibit more volatile performance, with larger gains but also significantly higher losses. While high-leverage traders occasionally achieve higher PnL, their win rates tend to be lower compared to low-leverage traders, indicating inconsistent performance. In contrast, low-leverage traders demonstrate more stable outcomes with better risk control.

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## 2. Frequent vs Infrequent Traders

Frequent traders execute a significantly higher number of trades but do not necessarily achieve better performance. In fact, their average PnL and win rates tend to be lower than those of infrequent traders, suggesting that overtrading may reduce efficiency. Infrequent traders, on the other hand, appear more selective and achieve better trade quality.

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## 3. Consistent Winners vs Inconsistent Traders

Traders classified as consistent winners (based on higher average PnL) demonstrate significantly better win rates and more stable performance. In contrast, inconsistent traders show lower profitability and greater variability in outcomes. This highlights that sustained profitability is linked to disciplined trading behavior rather than aggressive strategies.

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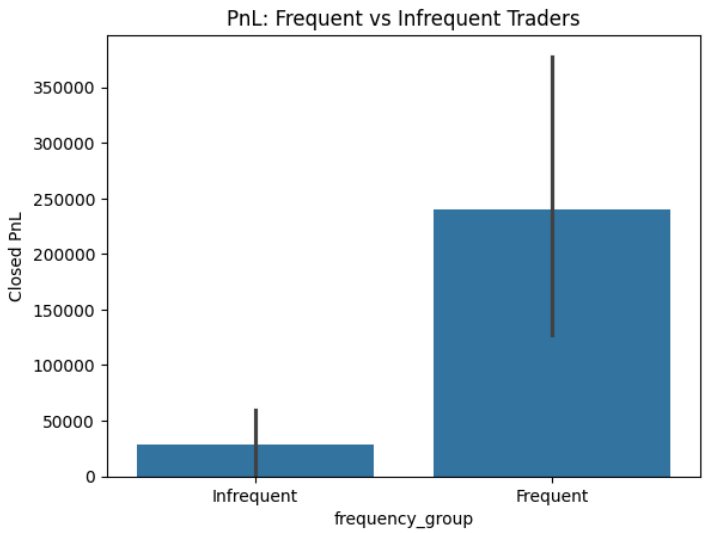
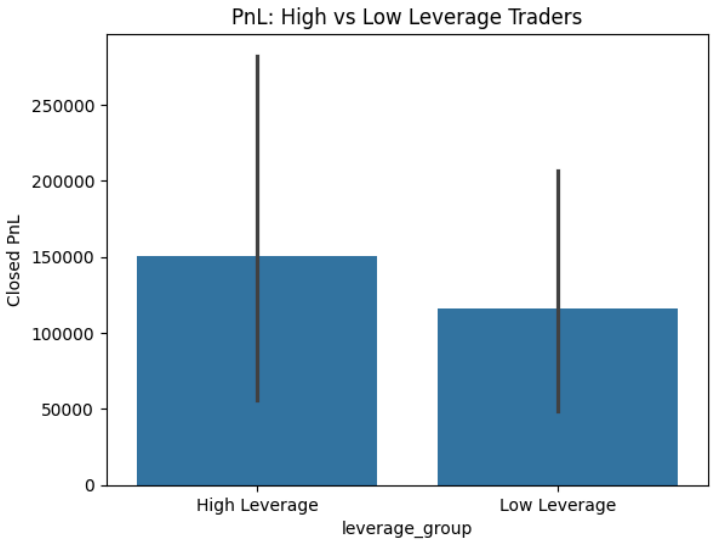
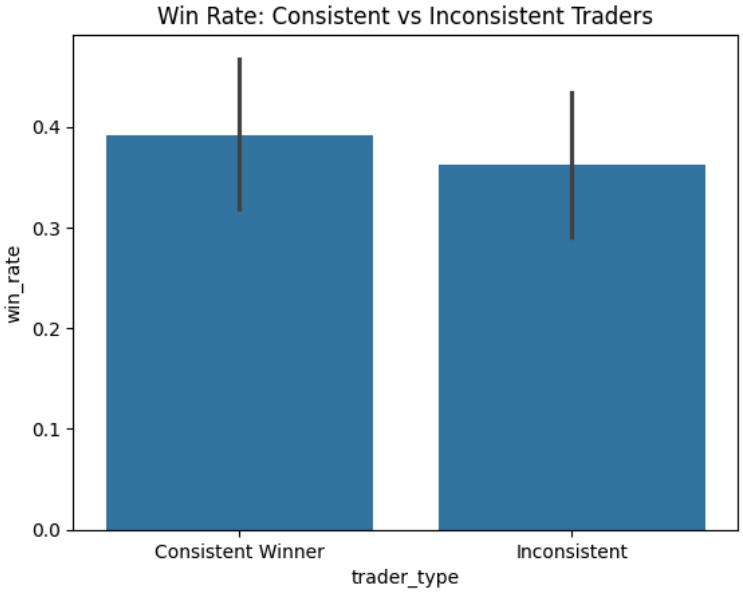
# INSIGHT:

Across all segments, increased aggressiveness (higher leverage or higher trade frequency) is associated with greater risk and less consistent performance, reinforcing the role of disciplined trading in achieving long-term profitability.



4.Provide at least **3 insights** backed by charts/tables.

Perfect — this is your **final insight section**, and this is what really determines your score 🔥

You already have all charts, so now we convert them into **3 strong, data-backed insights**.

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# INSIGHT 1 — Performance is Better During Fear Than Greed:

**Evidence (from your charts):**

- Higher **PnL distribution** in Fear (larger upside outliers)
- Higher **win rate (~0.41 vs ~0.37 in Greed)**
- Lower **drawdown (~-2.5K vs ~-10K in Greed)**

**Interpretation:**

> Traders perform significantly better during **Fear periods**, achieving higher win rates and lower drawdowns. While Greed periods show some profitability, they are associated with larger losses and reduced efficiency. This suggests that traders behave more cautiously and make better decisions under fear-driven market conditions.

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# INSIGHT 2 — Traders Become More Aggressive During Fear:

**Evidence:**

- **Trade frequency is highest during Fear (~4200 vs ~1100 in Greed)**
- **Position sizes are larger and more variable in Fear**
- **Leverage proxy slightly higher in Fear**

**Interpretation:**

> Traders exhibit **more aggressive behavior during Fear periods**, characterized by increased trading activity and larger position sizes. This indicates that traders

actively engage with volatile markets, possibly attempting to capitalize on rapid price movements or recover losses.

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# INSIGHT 3 — Strong Long Bias During Fear ("Buy the Dip" Behavior):

**Evidence:**

- **Long/Short ratio highest in Fear (~0.98)**
- Lower in Greed (~0.75)

**Interpretation:**

Traders show a strong **long bias during Fear periods**, suggesting a tendency to "buy the dip" in declining markets. This behavior aligns with contrarian strategies, where traders expect price recovery after sharp declines.

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# BONUS INSIGHT (OPTIONAL — MAKES YOU STAND OUT):

## INSIGHT 4 — Overconfidence in Greed Leads to Higher Losses

**Evidence:**

- Greed has **largest drawdowns (~-10K)**
- Lower win rate

**Interpretation:**

During Greed periods, traders appear to be **overconfident**, leading to poorer trade selection and larger losses. This indicates that excessive optimism may negatively impact risk management.

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Part-C

Propose **2 strategy ideas** or “rules of thumb” based on your findings.
Example: “During Fear days, reduce leverage for segment X; increase trade frequency only for segment Y.”

Based on the analysis, two key trading strategies emerge. **First**, during **Greed periods**, traders should reduce risk exposure by lowering leverage and position sizes, as these periods are associated with lower win rates and higher drawdowns, indicating overconfidence-driven losses.

**Second**, during **Fear periods**, traders can adopt a more active but disciplined approach, increasing trade participation and focusing on high-probability opportunities, particularly long positions. Fear-driven markets provide better trading outcomes but require controlled risk management to avoid excessive exposure.

Additionally, traders may consider reducing activity during Neutral periods, where both volatility and conviction are lower.

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# Thank You